



CALEB PENLEY END OF TERM PRESENTATION

First Term | A2 Process Engineering | Manager: Jeffrey David | Mentor: Tino Veitengruber



1. About Me
2. Major Projects
3. Minor Projects
4. Professional Development
5. Next Steps



About Me

- Grew up in PG County, Maryland
- Senior studying Mechanical Engineering at University of Maryland
- Interested in photography and sports
- My love for cars started when I changed my car's alternator with a friend as a junior in high school



Yes, I took that picture of the AMG GT3 in the top right.

MAJOR PROJECTS

Trailer Hitch Hand-Start Tool - Trim 1

PROBLEM:

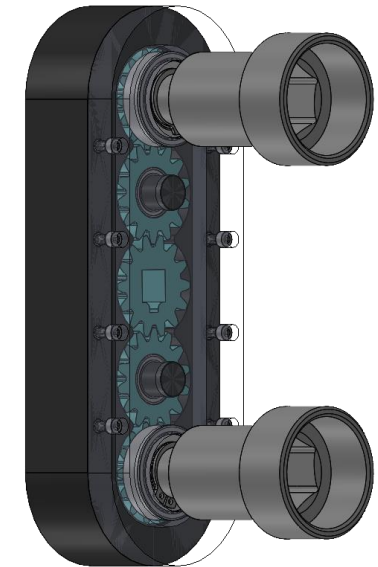
- Trailer hitch has fallen off on-line 2 times in past 4 months due to insufficient hand-starting on Trim 1
 - Posed a safety risk to TMs
- Hitch is hand-started on Trim 1, torqued on Final 1

ACTION:

- Designed and 3D printed prototype that:
 - Prevents safety incidents
 - Provides repeatability
 - Doesn't prematurely engage thread locker
- Mechanism works by using a sleeve on the socket that hits the hitch, the bolt screws itself out of the socket

RESULT:

- **Eliminated safety incidents**
- Reduced process speed from **7 to 3 seconds** (57%)
- Reduced price from **\$11,800 to \$3,600** (70%)
- Reduced lead time from **10 to 2 weeks** (80%)
- Handed design off to F&S to be manufactured
 - Durable material
 - Set up in Central Stores



CAN Tools New Process - Door Line

PROBLEM:

- New CAN tool for MOPF2 added 15 seconds per door to communicate with door's control unit

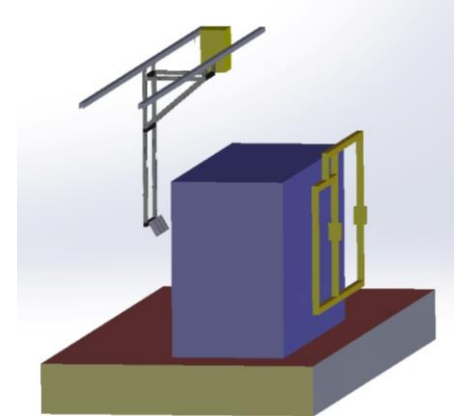
ACTION:

St. 10 (Lower windows)

- Designed flow rack concept to split process between St. 9 and St. 10
- Ran 4 trials
 - First 3 trials split the processes between St. 7-10
 - Final trial only restructured St. 10

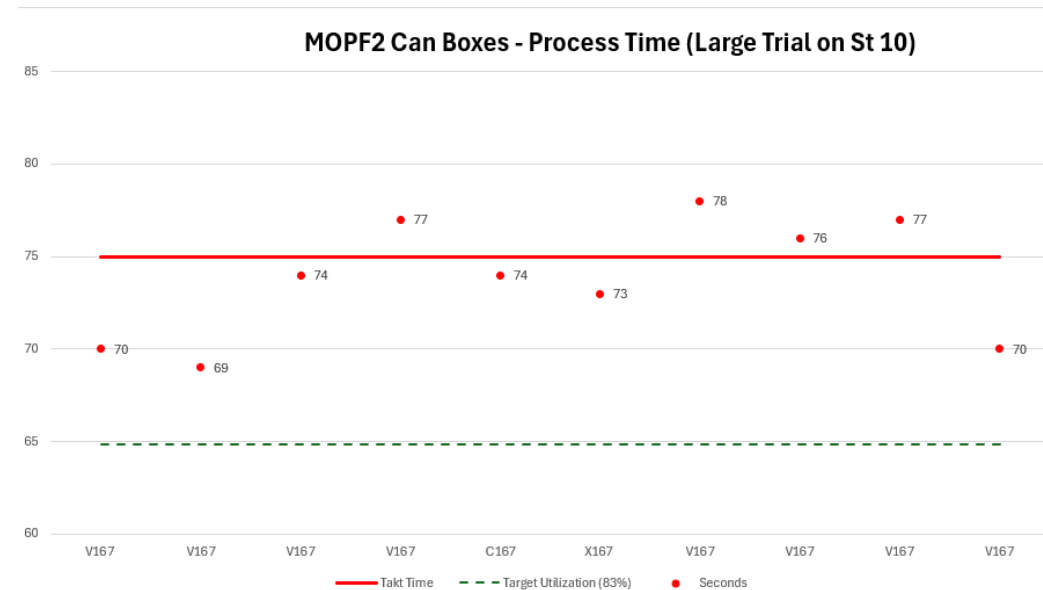
St. 17 (Raise windows)

- Designed rail system to implement CAN tools
- Attempted trials on PT1a
 - CAN tool not working, so waiting for PT1b for more trials



RESULT:

- Reduced utilization in St. 10 from **115% to 97%** by restructuring the process to eliminate the 15-second idle wait time
- Handed St. 17 off to SP, PE, and Production
- Personal development: cross-team functionality, adaptability, creativity, process changes, supplier communication



Hood Stops

ORIGINAL HOOD STOP



PROS:

- Good connection to hood
- Design provides place for TMs to place cables

CONS:

- Risk of falling out, causing downtime
- Wear on hood stop

IDEA FROM GERMANY



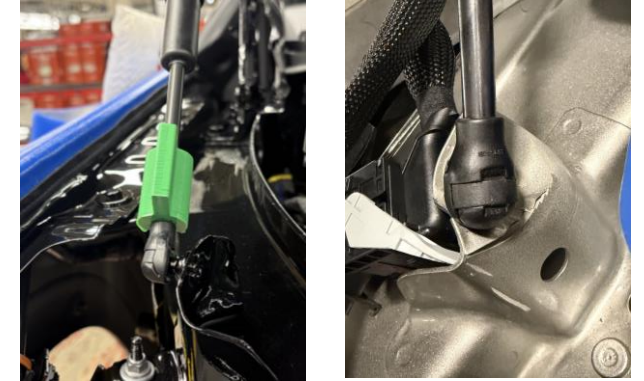
PROS:

- Connects to fender covers, won't fall
- Easy to manufacture (3D print)

CONS:

- Nowhere to hang cables
- Bad contact points on hood (not the same as in Germany)

STRUT SPACER



PROS:

- Simple concept
- Easy to manufacture (3D print)

CONS:

- Nowhere to hang cables
- Bends strut connection bracket

Project Start (CW 44)

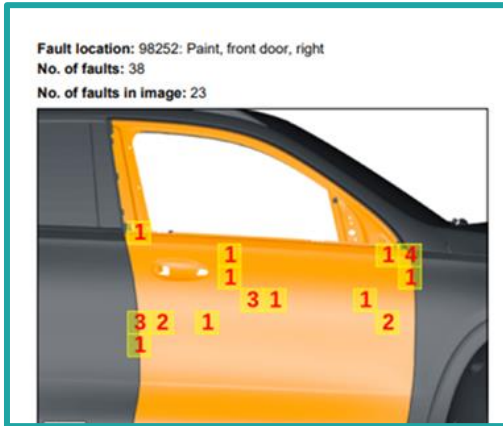
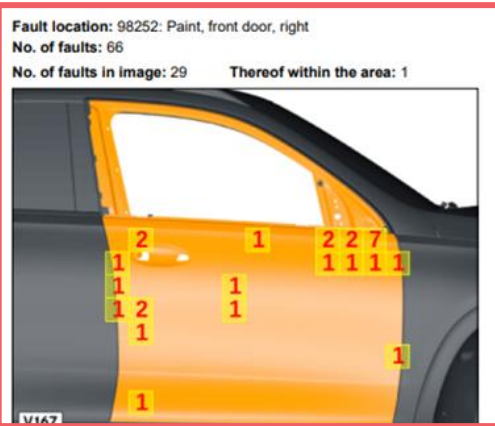
Iteration II Design +
Test (CW 47-48)

Brainstorm Other
Options (CW 49-50)

Iteration I Design + Test
(CW 44-47)

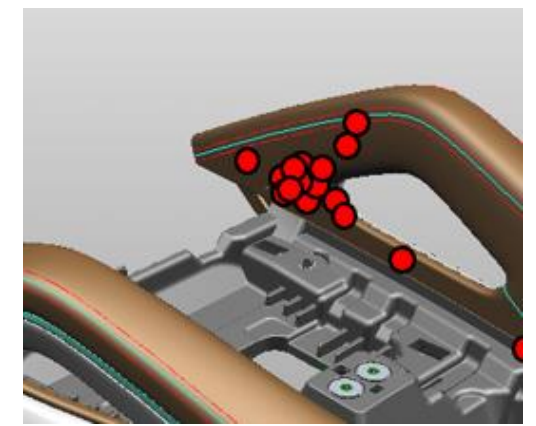
Strut Spacer Design +
Test (CW 49)

MINOR PROJECTS



- Decreased scratches from trim piece installation by 66%

** Data from 3 weeks before and 3 weeks after **



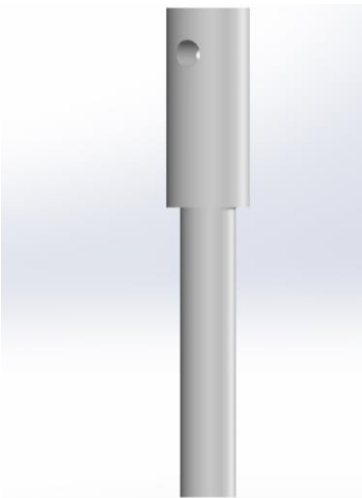
- Tested 3 different locations for holders, got feedback from TMs
- Helped reduce scratches and tears from sharp corner

Jigs



Carpet Jigs

- Keeps carpet in place on Trim 5 so that bolt holes are visible for TMs putting parts on car's interior
- Conforms to geometry of the car, magnets and bolt hold jig in place



6-week Trend - Coolant Hose “Not Connected Issues” from Trim 4

# Issues Before Jig	# Issues After Jig
82	32

Coolant Hose Connection Test Jig

- 61% reduction in “Not Connected” issues
- Likely reason for the 32 issues is TM not using jig (process not in SMP, nights not aware of new jig, 1st design missing a place to store jig)

PROFESSIONAL DEVELOPMENT

Professional Development



Cross-Team Functionality

Worked in teams with SP, Production, MOE, KVP, Quality, QRD, and RD.



Root-Cause Analysis

Found and solved process issues with PEs to increase torque FTC.



Leadership / Ownership

Set reasonable timelines, managed time well, reported to PEs and E4 before updates were requested.



Technical Skills

Learned Siemens NX, Engineering Client, MayCad, and KPIs. Improved on Solidworks, 3D printing, and tool operation.

NEXT STEPS

Mercedes-Benz



University of Maryland



Graduate from
University of Maryland

Spring '26

Summer '26

Fall '26

Spring '27

Full Time at
MBUSI?



MBUSI
2nd Term



MBUSI
3rd and Final Term

Mercedes-Benz

THANK YOU MBUSI

